

Loan Recovery Rate Prediction for Asset Reconstruction Company (ARC)

Model Selection and Approach

The objective of this assignment was to predict the **recovery_rate** for defaulted loans using historical loan and recovery-process information. Since the dataset consists of both numerical and categorical features with several missing values, I selected **CatBoost Regressor** as the primary model.

CatBoost was chosen because it naturally handles categorical variables, missing values, and nonlinear relationships while requiring minimal preprocessing. This makes it well suited for structured financial datasets where feature interactions are often complex.

To improve predictive performance, I performed feature engineering by creating the following derived features:

- Average FICO score (`fico_avg`)
- FICO score spread (`fico_spread`)
- Loan-to-income ratio
- Revolving balance-to-income ratio

To prevent target leakage, the following columns were excluded from model training:

- `recoveries`
- `collection_recovery_fee`
- `recovery_rate` (target variable)

Additionally, `payment_plan_status` was removed because it is unavailable in the test dataset and therefore would not be available during inference.

Evaluation

The dataset was split into **80% training** and **20% validation** using a fixed random seed for reproducibility.

Since this is a regression problem, multiple evaluation metrics were used instead of relying on a single score.

Metric	Value
MAE	0.0950
RMSE	0.1179
R ²	0.6291

MAE provides an intuitive measure of average prediction error, RMSE penalizes larger prediction errors more heavily, and R² measures the proportion of variance explained by the model. Using all three metrics provides a more comprehensive assessment of model performance.

Important Features

Feature importance obtained from the trained CatBoost model identified the following variables as the strongest predictors:

1. **recovery_status**
2. **days_past_due_at_default**
3. **Debt-to-Income Ratio (DTI)**

Other influential features included:

- Average FICO score
- FICO range
- Delinquencies in the past two years
- Collateral information
- Employment length

These features are consistent with domain intuition, as repayment behavior, borrower credit quality, delinquency history, and collateral characteristics directly influence loan recovery outcomes.

Considerations for Production Deployment

Before deploying this model in a real underwriting or recovery workflow, several additional steps would be necessary:

- Validate the model on larger and more diverse real-world datasets.
- Continuously monitor feature distributions and prediction quality to detect data drift and concept drift.
- Periodically retrain the model as borrower behavior and macroeconomic conditions change.
- Perform fairness and bias analysis to ensure consistent performance across borrower segments.
- Incorporate human review for high-value recovery decisions rather than relying solely on automated predictions.
- If predictions are required immediately after loan default, develop a separate model that excludes recovery-process variables that may not yet be available.

Overall, this solution emphasizes preventing target leakage, building an interpretable machine learning pipeline, and following practices that would support reliable deployment in a production environment.